

MD KASHIF ALAM

Robotics Software Engineer | ROS2 • Computer Vision • Embedded Systems • AI Integration

CONTACT

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TECHNICAL SUMMARY

I am a Robotics Engineer at IIT Patna with a long-standing passion for building autonomous systems. For over seven years, I have documented my evolution from custom electronics to high-level robotics on my platform, **Electrobotics AI**. I specialize in bridging the gap between embedded firmware and ROS2 navigation, focusing on creating hardware that interacts intelligently with its environment.

EDUCATION

- **Bachelor of Science (BS)**, Artificial Intelligence and CyberSecurity | IIT Patna
- **Computer Science (CS)** | Institute of Innovation (PW), Bangalore

TECHNICAL PROFICIENCY

- **Robotics Frameworks:** ROS2 (Humble), Nav2 Stack, MoveIt2, URDF Modeling, TF2
- **Programming Languages:** C++ (OOP), Python, MicroPython, Bash Scripting
- **Computer Vision & AI:** OpenCV, YOLO, MediaPipe, TensorFlow, SLAM
- **Embedded Systems:** ESP32, STM32, Raspberry Pi, Arduino, PCA9685
- **Control & Kinematics:** PID Tuning, Inverse Kinematics, Gait Dynamics, State Estimation
- **Hardware Communication:** UART (Serial), I2C, SPI, CAN Bus (Basics)
- **Simulation & Modeling:** Gazebo, RViz2, Fusion 360 (CAD), MATLAB/Simulink
- **Development Tools:** Linux (Ubuntu), Git/GitHub, VS Code, WSL2, Docker (Basics)

TECHNICAL EXPERIENCE

- **Quadruped Robotic Dog (Ongoing)**
 - Engineering an **autonomous quadruped robot** capable of walking, running, and turning, with future integration of **AI-based navigation, mapping, and localization**.
- **Ornithopter (Ongoing)**
 - Engineered a **gearbox-driven flapping-wing mechanism** replicating **avian flight**, and progressing the **control system** for **stable, semi-autonomous aerial surveillance**.
- **AI-Powered Gesture-Controlled Robotic Hand**
 - Developed a robotic hand controlled via **OpenCV** and **Mediapipe**.
 - Integrated **servo motors** with **real-time vision processing** for accurate hand movements.
- **Sign-Controlled Light**
 - Implemented **gesture recognition** to control lighting — *Thumbs Up → Light ON, Thumbs Down → Light OFF*.
- **AI-Powered Mentor**
 - Built an AI mentor application with a **floating interface** to capture screen or voice input, process via **AI APIs**, and respond intelligently in **voice or text**.
- **Robotic Arm**
 - Engineered a **robotic arm** capable of **precise 3D point movement** using multiple **servo motors** for accuracy and smooth control.
- **Surveillance Drone**
 - Designed and built a drone equipped with **real-time video streaming** for **aerial monitoring and surveillance**.